

The Mr Tickle Screen

A reductio ad absurdum against physical closure from bare observer screens

No Semantic Alchemy, No Bulk From Bare Boundary, and physical non-definability in Observer Patch Holography

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Authorship intentionally withheld in the public package

Central result

Assume that a source-only observer-screen architecture is sufficient to make its formal screen a physically instantiated holographic boundary and to select a physical bulk. Then any source-isomorphic presentation—including the “Mr Tickle Screen” constructed below—inherits the same alleged physical status. But source mathematics cannot constrain an external physical- realization relation: the same source model admits an expansion in which that relation is empty. Moreover, a bare twelve-port incidence structure does not determine the boundary quantum theory required by a holographic dictionary. Therefore physical closure cannot be obtained from A1–A3, nor from any purely source-internal enrichment of them. A successful future closure must supply independently grounded physical boundary dynamics and a reconstruction dictionary; once supplied, those are additional load-bearing physical structure rather than consequences of the bare screen alone.

Frozen audit baseline. FloatingPragma/observer-patch-holography, commit 40b13bb4bc56261f84fd515701916c388c42fe83.

Live public snapshot checked. Main-branch commit 05207466cf63461c683bf1b8543d7c4a5feea4fa (26 August 2026).

Priority statement. The source/realization countermodel was documented in a hash-anchored prior audit dated 2 August 2026. The present article generalizes that witness into a source-only physical-closure impossibility theorem, a holographic typing obstruction, and a Lean 4 proof kernel.

Scope. Internal source theorems may remain correct. The claim under attack is that source formalism, observer consensus, or theorem volume can by themselves establish physical instantiation or a unique physical bulk.

Abstract

We present a *reductio ad absurdum* against deriving physical ontology from a bare observer-screen formalism. The *reductio* is intentionally simple. Let a source theory characterize an abstract screen, records, overlaps, agreement, response algebras, fixed points and an information-projection rule. Assume, for contradiction, that satisfaction of those source conditions is sufficient to establish that the screen is physically instantiated and that a physical bulk is thereby selected. Because the source theory is presentation- invariant, an isomorphic source model can be displayed on an arbitrary pictorial carrier. We choose an original long-armed orange cartoon schematic, the “Mr Tickle Screen”, carrying the same formal twelve-port source data. If source structure alone established physicality, the cartoon presentation would inherit that status. The absurd conclusion reveals the missing type: physical instantiation is not a source-language property. Formally, a satisfiable source theory admits an expansion with an empty external realization relation; therefore no source-only enrichment can force external physical realization. We call this result *No Semantic Alchemy*.

A second, independent obstruction comes from holography. AdS/CFT and HKLL do not reconstruct a bulk from bare boundary incidence. They act on a specified boundary quantum theory with operator algebra, states, dynamics and correlators, together with a duality/reconstruction dictionary. The same finite twelve-site carrier can support inequivalent Hamiltonians and correlation structures, so the bare screen cannot uniquely select the boundary theory that a holographic reconstruction requires. We formalize the result as *No Bulk From Bare Boundary*: if two boundary theories share the same bare screen but have different reconstruction outputs, no bulk assignment can factor through the bare screen alone. Combining these results with the previous same-reduct physical non-definability theorem yields a strict closure dichotomy for Observer Patch Holography (OPH): remain source-only and physical closure is impossible; or introduce independently grounded physical boundary dynamics, calibration and a holographic dictionary, in which case those objects are additional load-bearing physical structure and the closure is no longer a consequence of A1–A3 alone. The theorem is independent of the number of machine-checked source consequences.

One-sentence reductio

If the same formal criteria that supposedly make an OPH screen physically fundamental can be carried unchanged by the Mr Tickle Screen, then either the cartoon has become a physical holographic boundary or those criteria were never sufficient to establish physicality. Formal annotation does not instantiate its referent, so the second horn follows.

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1 Introduction: why the absurdity is the argument

The central error audited here is a type error between *characterizing a formal object* and *establishing that nature instantiates it*. The error can hide for a long time when the formal vocabulary uses physical nouns. Calling mathematical objects “observers”, “screens”, “records”, “currents”, “repair”, “meaning”, “horizons” or “bulk” does not turn those objects into laboratory systems. Nor does adding more internally valid theorems. Formal consistency is a relation among statements or structures; physical occurrence is a relation between a mathematical representation and the world.

The point becomes obvious under a deliberately absurd presentation. Take a source model satisfying every relevant formal condition and draw its carrier on a long-armed orange cartoon figure. Call this the *Mr Tickle Screen*. The artwork is irrelevant: the abstract source model is unchanged. If one claims that the source conditions themselves are sufficient to make the screen physically instantiated, then the same implication applies to this presentation. The conclusion is absurd not because cartoons cannot support mathematics, but because mathematics attached to a cartoon does not cause the cartoon to occur in nature. The only information used to reject the physical Mr Tickle Screen is information external to the source formalism—precisely the missing physical-realization content.

This reductio is then made exact in three steps. First, standard reduct/ expansion machinery shows that a satisfiable source theory cannot force an external realization relation: the empty relation is always a same-source expansion unless physical realization has been imported into the premises. Second, factorization theory shows that a bare screen cannot determine a boundary quantum theory when two inequivalent boundary dynamics live on the same screen. Third, the holographic dictionary is typed from a specified boundary theory to a bulk description; a screen is not the required input. Thus no source-only theorem can jump directly from incidence/consensus to a physical bulk.

The target is not a caricature of holography. In AdS/CFT, the boundary side is a specified conformal quantum field theory and the bulk side is a specified gravity/string theory. Maldacena’s proposal and Witten’s generating-functional dictionary relate those two theories [5, 6]. HKLL then represents local bulk fields by nonlocal boundary operators in the semiclassical regime [7]; modern bulk reconstruction can also be formulated in quantum-error-correcting language [8]. None of this says that drawing a boundary creates a bulk. It says that, given a physical boundary theory and a valid duality dictionary, bulk observables can be reconstructed from boundary data.

1.1 What is borrowed and what is new

The machinery is standard: reducts and expansions, model-theoretic satisfaction, Beth definability, forgetful functors and factorization, observational equivalence, gauge-invariant readouts, information projection, and the established AdS/CFT/HKLL reconstruction architecture [9, 10, 11, 12, 13, 14, 5, 6, 7, 8].

The new contribution is the composition of these tools into an explicit reductio and a closure theorem. The argument does not merely say that a physical bridge is “open”. It proves that *no bridge whose content remains entirely inside the source language can perform the missing physical work*. If a later bridge supplies independent boundary dynamics, laboratory calibration or a holographic dictionary, then it may be scientifically valuable, but it is additional physical structure and must be counted as such.

2 The Mr Tickle Screen reductio

2.1 Construction

Let T_S denote a source theory and let $M \models T_S$. In the OPH application, M contains the twelve-port boundary packet, overlap/seam structure, record and readback interfaces, response algebra, agreement maps, and A3 state-selection data. Let P denote a source presentation. A presentation change is admissible when it preserves every source object and map that the theory treats as protected.

The Mr Tickle Screen is not a competing physical theory. It is the same abstract source model displayed on a deliberately ridiculous pictorial carrier. The point is to remove any accidental persuasive force contributed by words such as “screen” or by a diagram that visually resembles a scientific boundary.

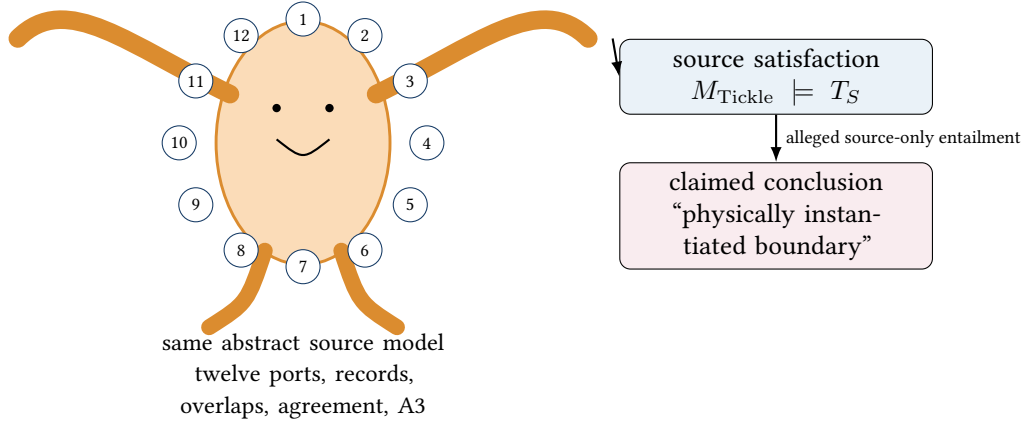


Figure 1: The Mr Tickle Screen reduction. The cartoon is an original schematic used for criticism/parody; it does not reproduce published Mr. Men artwork. The source-theoretic point is independent of appearance: a presentation can carry the same abstract mathematics without acquiring physical instantiation. Only the port locations are sketched; the full abstract 30-edge/20-face packet is unchanged and is not redrawn on the cartoon.

2.2 Reductio ad absurdum

Assume for contradiction that source satisfaction is sufficient for physical instantiation:

$$M \models T_S \implies \text{PhysReal}(M). \quad (1)$$

If M_{Tickle} is source-isomorphic to M , then every presentation-invariant source theorem has the same truth value on both. The assumption (1) therefore yields

$$\text{PhysReal}(M_{\text{Tickle}}).$$

But no pictorial recoding, no matter how mathematically elaborate, creates a new physical object. The rejection of that conclusion must appeal to facts outside the source presentation. Therefore the source structure was not sufficient for physical instantiation in the first place.

Reductio 2.1 (Mr Tickle Screen). *Let T_S be presentation-invariant and satisfiable. If physical instantiation is not itself fixed by the source language, then source satisfaction cannot entail physical instantiation. In particular, replacing a scientific-looking presentation by the Mr Tickle Screen cannot change any source theorem but also cannot create a physical boundary. Hence any rule that promotes the former but not the latter must consume additional non-source information.*

Why the joke is mathematically useful

The reductio does not argue that a fictional character is a model of physics. It argues that *formal annotation is not an ontological creation operator*. A cartoon can carry a graph, an algebra, a probability law and an arbitrarily large theorem library. Those properties remain properties of the formal representation until an independently grounded physical-realization relation is established.

3 No Semantic Alchemy

Let L_S be the source language and let T_S be a satisfiable theory in that language. Introduce a distinct relation

$$\mathcal{R} \subseteq \text{Mod}(T_S) \times \mathcal{W},$$

where $\mathcal{R}(M, w)$ means that source model M is physically instantiated by world/system w . The relation is intentionally external to L_S : it expresses the representation-to-world fact whose existence is at issue.

Theorem 3.1 (No Semantic Alchemy). *If T_S is satisfiable and contains no condition that fixes the external relation*

$\mathcal{R}$, then

$$T_S \not\models \exists w \mathcal{R}(M, w).$$

More strongly, for any source-only enrichment $T_S^+ = T_S \cup \Delta$ that remains satisfiable and still does not constrain $\mathcal{R}$,

$$T_S^+ \not\models \exists w \mathcal{R}(M, w).$$

Proof. Choose a source model $M \models T_S^+$. Form two expansions with the same source reduct. In the first, set $\mathcal{R} = \emptyset$; in the second, choose any nonempty realization relation. Every L_S -sentence has the same truth value in the two expansions because their source reducts are identical. The first expansion falsifies $\exists w \mathcal{R}(M, w)$. Therefore no source-only sentence can entail that existential claim. $\square$

The result is not changed by adding an internal predicate named $\text{Physical}(x)$. Such a predicate is still interpreted inside the source model. One can satisfy $\exists x \text{Physical}(x)$ while leaving the external realization relation empty. Naming a mathematical object “physical” is not a proof that nature instantiates it.

This closes the “one more axiom” escape. If the new axiom remains a source-internal structural condition, the empty-realization expansion survives. If the new axiom genuinely fixes physical realization by importing laboratory facts, calibrated dynamics, a boundary QFT or a duality dictionary, then the new content is an additional physical premise. Either way, it cannot be credited to the original source structure alone.

4 No Bulk From Bare Boundary

4.1 The holographic type signature

A bare finite screen and a holographic boundary theory are not objects of the same type. Schematically,

$$\text{Screen} \neq \text{BoundaryQFT}.$$

A boundary quantum theory contains, at minimum, a state/Hilbert structure, an observable/operator algebra, dynamics, and correlation data. A holographic dictionary then relates that specified boundary theory to a bulk theory. In AdS/CFT, Maldacena’s duality relates a specific CFT sector to a specific gravity/string-theory sector [5]; Witten’s formulation relates CFT generating functionals to bulk gravitational boundary conditions [6]. HKLL is further downstream: it represents local bulk operators by smeared, nonlocal boundary operators in the semiclassical regime [7].

Thus the typed chain is

$$\text{Screen} \xrightarrow{L} \text{BoundaryQFT} \xrightarrow{\mathcal{D}} \text{BulkTheory} \xrightarrow{O} \text{PhysicalReadout}. \quad (2)$$

The first lift L and the dictionary $\mathcal{D}$ are load-bearing. A bare incidence packet is not an HKLL input.

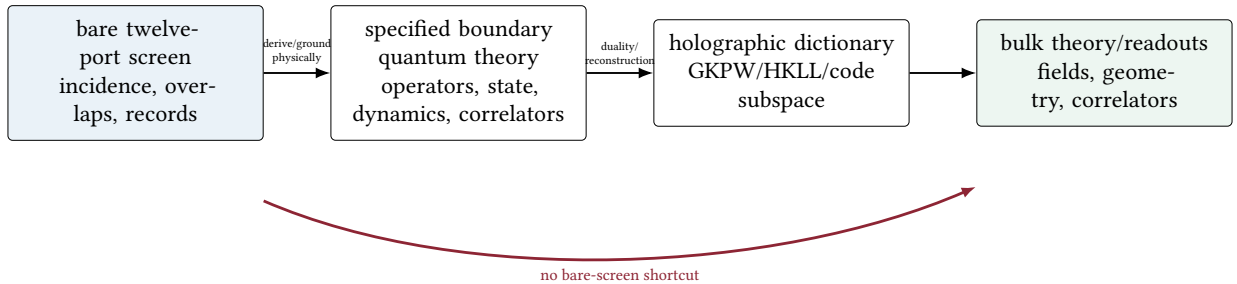


Figure 2: The holographic typing obstruction. HKLL is downstream of a specified boundary quantum theory and a reconstruction dictionary; it is not a map from a finite incidence drawing directly to a physical bulk.

4.2 The same twelve-point carrier supports different physics

The failure already occurs before holography. Put twelve qubits on the same icosahedral incidence graph K_{12} , with

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes 12}.$$

Consider two Hamiltonians on exactly that carrier:

$$H_0 = 0, \quad H_J = J \sum_{\langle ij \rangle \in E(K_{12})} Z_i Z_j, \quad J \neq 0.$$

The screen, port count, edge count and incidence are identical. The dynamics, spectra, time evolution and generic correlation functions are not. Therefore the forgetful map

$$U : \text{BoundaryTheory} \rightarrow \text{BareScreen}$$

has a non-singleton fibre over the same twelve-port screen.

Theorem 4.1 (No boundary theory from bare screen). *If $B_1 \neq B_2$ and $U(B_1) = U(B_2)$, then there is no selector $S : \text{BareScreen} \rightarrow \text{BoundaryTheory}$ satisfying $S(U(B)) = B$ for every boundary theory B .*

Proof. If such S existed,

$$B_1 = S(U(B_1)) = S(U(B_2)) = B_2,$$

contradicting $B_1 \neq B_2$. □

Theorem 4.2 (No Bulk From Bare Boundary). *Let $H : \text{BoundaryTheory} \rightarrow \text{BulkTheory}$ be any bulk assignment/reconstruction on a class of boundary theories. If*

$$U(B_1) = U(B_2) \quad \text{but} \quad H(B_1) \neq H(B_2),$$

then no map $\hat{H} : \text{BareScreen} \rightarrow \text{BulkTheory}$ can satisfy $H = \hat{H} \circ U$.

Proof. Factorization would give

$$H(B_1) = \hat{H}(U(B_1)) = \hat{H}(U(B_2)) = H(B_2),$$

a contradiction. □

The theorem is deliberately general. It does not assert that every twelve-site spin system has an AdS dual. It establishes the logically prior fact that a bare screen is insufficient even to select the boundary dynamics that a reconstruction dictionary would consume. Holography cannot repair missing input data by vocabulary.

4.3 Why HKLL sharpens rather than weakens the reduction

HKLL reconstructs a local bulk field using boundary operators in a specified AdS/CFT setting. Schematically,

$$\phi_{\text{bulk}}(X) = \int_{\partial \text{AdS}} K(X | x) \mathcal{O}(x) dx.$$

The boundary operator $\mathcal{O}$, its state/correlators and the smearing kernel are not determined by the statement “there are twelve points.” The kernel is tied to bulk/boundary dynamics and geometry; the boundary operators belong to a specified quantum theory [7]. In the quantum-error-correcting formulation, bulk operators are logical information encoded in a boundary code structure, again presupposing the code, state space and operator algebra [8].

The relevant distinction is therefore not “real versus mathematical” in a naive sense. Both sides of a duality are mathematical descriptions. The physical distinction is that the AdS/CFT dictionary relates already specified, physically interpreted quantum/gravitational degrees of freedom within a candidate physical duality. A bare cartoon or incidence graph has no such status merely because one calls it a holographic screen.

5 Physical closure cannot be source-only

It is useful to separate six logically different closure levels:

C1. Formal consistency: the source axioms have models.

C2. Source determination: desired mathematical objects are unique inside the source language.

C3. Physical instantiation: the source represents an actually realized physical system.

C4. Boundary-theory closure: the relevant physical boundary operator algebra, state and dynamics are fixed.

C5. Holographic closure: a valid boundary-to-bulk dictionary/code subspace is fixed and reconstructs the claimed bulk observables.

C6. Empirical closure: reconstructed observables are calibrated to and survive independent experiment.

Source-internal theorem proving can move within C1–C2. [Theorem 3.1](#) blocks a purely logical jump from C2 to C3. [Theorem 4.1](#) blocks a bare-screen jump to C4. [Theorem 4.2](#) blocks a bare-screen shortcut to C5. Experiment and calibration are required for C6.

No-escape theorem 5.1 (Source-only physical-closure impossibility). *Let T_S be any satisfiable source theory whose extensions remain entirely source-internal. Suppose physical closure requires (i) nonempty external physical realization, (ii) a physically specified boundary theory, and (iii) a bulk reconstruction whose physical readouts are unique. Then no sequence of source-only enrichments of T_S can establish physical closure. A successful closure must introduce independently grounded physical structure at or before the realization/boundary-theory stage.*

The theorem gives an exhaustive dichotomy for a programme such as OPH:

Remain source-only	$\implies$ physical closure is impossible,
add physical boundary/dictionary structure	$\implies$ closure is not from A1–A3 alone.

There is no theorem-count escape. More source consequences can characterize the formal screen with increasing precision, but they cannot create the external realization relation or supply a missing boundary quantum theory.

6 Version lock and target of the audit

6.1 Frozen OPH basis

The public target is the OPH repository at commit

40b13bb4bc56261f84fd515701916c388c42fe83,

committed on 26 August 2026. The reader-facing normative basis is docs/AXIOM_REFERENCE.md; the repository also names a machine-readable axiom registry. At this commit OPH declares three core axioms:

- A1.** an oriented twelve-port observer-screen architecture with a finite carrier federation, seams, records, readback, repair interfaces, an icosahedral boundary packet, an S^2 refinement support and a finite compact unitary response algebra;
- A2.** natural agreement of accepted observer-accessible data and operational meaning across overlaps, rechartings, transports and refinements;
- A3.** a conditional maximum-randomness/information-projection rule selecting a state inside a fixed feasible family relative to declared reference data.

The same reference expressly separates these source axioms from laboratory identification, global compact-group selection, field lists, multiplicities, model-space selection, couplings and other physical attachments. It also states that an exact result inside one named realization does not establish that the axioms select that realization [2].

6.2 Live public-claim snapshot

The logical theorems below are version-independent, but the current public claim surface remains relevant. At the frozen main-branch snapshot 05207466cf63461c683bf1b8543d7c4a5fee4fa, OPH describes itself as a zero-dial theory-of-everything programme in which observers are primary, objective reality is emergent, and overlap repair on a holographic screen is used as the route toward quantum theory, spacetime, gravity, gauge structure, matter and constants [3]. The same README advertises more than 8800 Lean theorems while also acknowledging that the physical attachment of its fixed-point closure remains open. This juxtaposition is exactly what the present theorem separates: source-proof volume and physical establishment are different quantities.

An archived explanatory chapter is even more explicit about the intended holographic move: it contrasts AdS/CFT’s specific boundary CFT with an OPH observer-horizon programme in which no global boundary CFT is required and the bulk is said to emerge from agreement/consistency of overlapping observer cuts [4]. That text is historical rather than the normative axiom registry; it is cited only to identify the physical intuition that [section 4](#) audits.

6.3 Claim under audit

The theorem does not require OPH to make one exact sentence in every public surface. It audits the following general claim schema:

$$A1\text{--}A3 \text{ and their source consequences uniquely recover physical structure } C. \quad (3)$$

Here C may be a physical gauge interaction, matter content, gravity, spacetime, coupling, measured constant, particle spectrum, laboratory current or any other experimentally meaningful object.

If OPH labels a result only as a conditional interface, named realization or physical identification, the no-go does not contradict that narrower status. It instead proves why the narrower status is required. The result directly blocks only the stronger wording in [equation \(3\)](#) unless a unique realization theorem is supplied.

6.4 Why version locking is non-negotiable

A rapidly changing theory can answer a countermodel by adding premises. That may improve the new theory, but it cannot alter what followed from the earlier one. We therefore write T_h for the theory at repository hash h . Every entailment statement below has the form

$$T_h \models C \quad \text{or} \quad T_h \not\models C.$$

A later $T_{h'} = T_h \cup \Delta$ is audited separately. This prevents a new premise from being presented retroactively as an old theorem.

7 Formal setting

7.1 Source and physical languages

Let L_{src} be a signature containing exactly the source objects and maps of a theory. For the OPH application these include the frozen A1–A3 objects: patches, ports, incidence, algebras, records, restriction maps, accepted updates, response representations, interpretation functors, feasible states, reference families and the information-projection rule.

Let $L_{\text{phys}} \supseteq L_{\text{src}}$ extend the source language by physical objects needed to express empirical claims. Depending on the branch, these may include:

spacetime events and metrics, Hamiltonians, stress-energy, fields,
global gauge groups, matter representations, couplings, detector maps,
scattering amplitudes, spectra, clocks and laboratory records.

A physical model P has a source reduct

$$U(P) = P \setminus L_{\text{src}},$$

obtained by forgetting the additional physical structure. In model-theoretic language, P is an L_{phys} -expansion of $U(P)$. Categorically, U is a forgetful functor

$$U : \text{Phys} \longrightarrow \text{Src}.$$

Definition 7.1 (Realization fibre). For a source model $M \in \text{Src}$, the realization fibre is

$$\text{Real}(M) = U^{-1}(M),$$

the class or category of admissible physical expansions whose source reduct is M .

7.2 Physical equivalence

Let $\mathcal{O}$ be the declared family of physical readouts. Each $O \in \mathcal{O}$ maps a physical model to a calibrated outcome in an outcome space Y_O .

Definition 7.2 (Observational equivalence).

$$P \sim_{\mathcal{O}} Q \iff \forall O \in \mathcal{O}, O(P) = O(Q).$$

This is deliberately stronger than identity of source-language answers. A physical distinction is active when at least one admitted physical readout separates the models.

Definition 7.3 (Physical determination). The source theory physically determines its realization relative to $\mathcal{O}$ when every realization fibre is observationally singleton:

$$\forall M \models T_{\text{src}}, \quad \forall P, Q \in \text{Real}(M), \quad P \sim_{\mathcal{O}} Q.$$

This definition allows many mathematical presentations of the same physics. It asks only that the source determine one observational equivalence class. It therefore cannot be defeated by harmless notation changes, coordinate changes or genuine gauge redundancy.

7.3 Gauge equivalence

Let $P \sim_g Q$ denote gauge equivalence. The defining physical requirement is

$$P \sim_g Q \implies P \sim_{\mathcal{O}} Q. \tag{4}$$

If a readout is gauge invariant and differs, the two models are not related by gauge. This simple condition closes the claim that a counterexample is merely a redescription.

8 Borrowed machinery

8.1 Reducts and expansions

If two L_{phys} -structures have the same L_{src} -reduct, every L_{src} -sentence has the same truth value in both. This is not a new physical principle. It is the definition of reduct and satisfaction. It means that no amount of proof performed solely in L_{src} can distinguish the two expansions.

8.2 Implicit and explicit definability

A new physical relation R is *implicitly definable* by T_{src} if no two models of the extended theory can agree on the entire source reduct and disagree on R . In first-order logic, Beth’s definability theorem says that implicit definability entails explicit definability: there exists a source-language formula φ_R such that

$$R(\mathbf{x}) \longleftrightarrow \varphi_R(\mathbf{x})$$

in every model of the theory [9, 10].

Beth’s theorem is used here as a demand, not as the sole proof. If a proponent claims that physical meaning is uniquely “implicit” in the source architecture, then in the first-order regime an explicit definition should exist. If the formalism is higher-order or categorical, the direct realization-fibre theorem below still applies without Beth.

8.3 Information projection

An information projection has the form

$$\rho_* \in \arg \min_{\rho \in \mathcal{K}} D(\rho \| \tau),$$

where $\mathcal{K}$ is a fixed feasible set and τ a fixed reference state. Standard I -projection theory studies existence, uniqueness and geometry inside that declared space [12, 13]. It does not, without further structure, select among different field lists, spacetimes, Hamiltonians or ontology classes. The distinction is the familiar one between state inference within a model and model-class identification.

8.4 Categorical form

The model-theoretic condition has a direct categorical translation. The source determines physical reality exactly when every fibre of $U : \text{Phys} \rightarrow \text{Src}$ is contractible after quotienting by observational equivalence. A fibre containing two observationally separated objects is not contractible. Adding more arrows or theorems inside the base category Src does not remove that multiplicity; one must constrain the lift.

9 The ten-theorem no-wiggle suite

9.1 T1: versioned entailment

Theorem 9.1 (Version-lock theorem). *Let T_h be the theory determined by a frozen signature, axiom set and repository hash h . Whether $T_h \models C$ is independent of premises added only in a later theory $T_{h'} = T_h \cup \Delta$.*

Proof. Entailment from T_h is quantified over models of T_h . Restricting to the smaller class of models satisfying Δ may make C true in $T_{h'}$, but it does not remove countermodels of T_h from the historical model class of T_h . Hence a later proof from $T_{h'}$ is not a proof from T_h . $\square$

9.2 T2: source-language indiscernibility

Lemma 9.2 (Reduct indiscernibility). *If*

$$P \upharpoonright L_{\text{src}} \cong Q \upharpoonright L_{\text{src}},$$

then every source-language sentence σ has the same truth value in P and Q .

Proof. Satisfaction of an L_{src} -sentence depends only on the interpretation of symbols in L_{src} . Isomorphic reducts preserve satisfaction. $\square$

Consequence

A proof assistant can certify arbitrarily many L_{src} -theorems without selecting between P and Q . The issue is not proof quantity; it is whether the physical lift is determined.

9.3 T3: two-expansion non-uniqueness

Theorem 9.3 (Two-expansion theorem). *Let $M \models T_{\text{src}}$. If $P, Q \in \text{Real}(M)$ and $P \not\sim_{\mathcal{O}} Q$, then T_{src} does not physically determine its realization.*

Proof. Physical determination requires every pair in $\text{Real}(M)$ to be observationally equivalent. The pair P, Q violates that universal condition. $\square$

9.4 T4: observable separation

Theorem 9.4 (Observable-witness theorem). *If there exists $O \in \mathcal{O}$ for which $O(P) \neq O(Q)$, then $P \not\sim_{\mathcal{O}} Q$.*

Proof. Observational equivalence requires equality for every member of $\mathcal{O}$. One counterexample readout negates the universal statement. $\square$

This apparently trivial theorem is load bearing. It prevents a response from moving the disagreement into an undefined metaphysical category. The witness is a physical record: a cross section, spectrum, clock comparison, current, Wilson line, detector count, lensing map or other calibrated outcome.

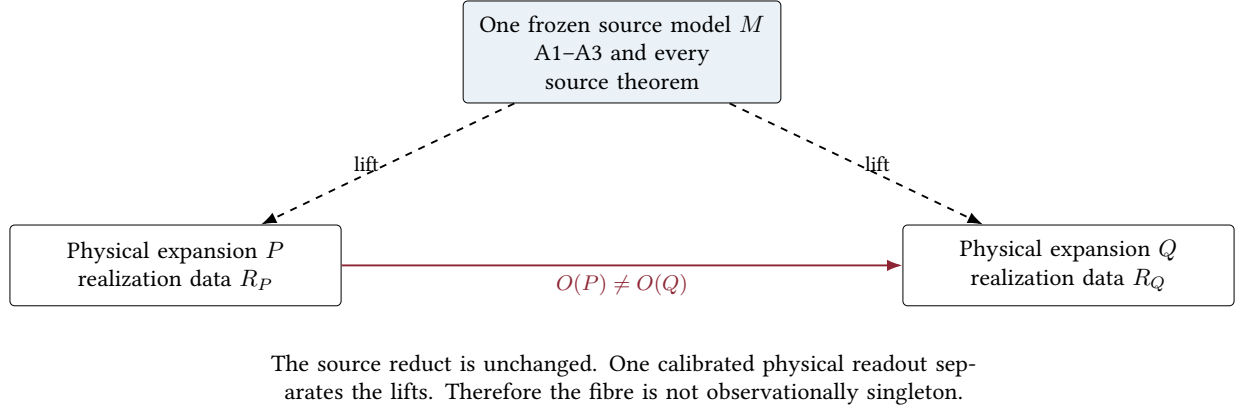


Figure 3: The universal counterexample pattern. It is invariant under changes of notation and does not challenge source-internal theorem validity.

9.5 T5: no gauge escape

Theorem 9.5 (Gauge-escape no-go). *Assume gauge equivalence preserves every admitted readout as in equation (4). If $O(P) \neq O(Q)$ for a gauge-invariant O , then $P \not\sim_g Q$.*

Proof. If $P \sim_g Q$, equation (4) gives $O(P) = O(Q)$, contradicting the witness. $\square$

Thus the counterexample must be answered physically, not by calling the two models different presentations.

9.6 T6: Beth’s explicit-definition demand

Corollary 9.6 (Beth fork). *In a first-order formalization, a physical relation that is uniquely implicitly determined by T_{src} must be explicitly definable in L_{src} . Therefore one of the following holds:*

- (i) *an explicit source-language definition is supplied;*
- (ii) *the relation is additional primitive structure;*
- (iii) *unique determination fails.*

Proof. Beth’s definability theorem gives (i) whenever implicit definability genuinely holds. If no such definition is part of the theory and the physical relation is nevertheless included, it is primitive, giving (ii). Otherwise there are models agreeing on the source reduct and differing on the relation, giving (iii). $\square$

The corollary blocks the phrase “the physical meaning is implicit” from serving as a proof. Implicit uniqueness is itself a mathematical claim with a definability consequence.

9.7 T7: information-projection blindness

Theorem 9.7 (Source-blind optimization theorem). *Let $U : \text{Phys} \rightarrow \text{Src}$ be the source reduct. Suppose the feasible predicate and objective factor through U :*

$$K(P) = K_0(U(P)), \quad J(P) = J_0(U(P)).$$

If $U(P) = U(Q)$, then

$$K(P) \iff K(Q), \quad J(P) = J(Q).$$

Consequently, a minimization rule using only K and J cannot uniquely select one of two distinct same-reduct lifts.

Proof. Factorization gives

$$K(P) = K_0(U(P)) = K_0(U(Q)) = K(Q)$$

and the identical calculation for J . If P is feasible and minimizes J , then Q is feasible with the same objective value and is also a minimizer. Uniqueness fails unless the rule uses information not factoring through the source reduct. $\square$

Application to maximum randomness

A3 may uniquely select a state in a fixed OPH source feasible space. It cannot select a unique Hamiltonian, field list, physical interpretation, global gauge form or laboratory map when those structures are absent from that source space. Adding them to the feasible space or reference measure makes them inputs to the selection problem.

9.8 T8: the interpretation-functor fork

Theorem 9.8 (Interpretation-functor theorem). *Let a source model contain an interpretation map $\mathcal{J} : \text{Data} \rightarrow \text{Meaning}$. For a physical identification encoded through $\mathcal{J}$, at least one of the following holds:*

- (a) $\mathcal{J}$ is fixed with the required empirical semantics, in which case that semantics is source data;
- (b) multiple admissible $\mathcal{J}$'s survive, in which case physical meaning is underdetermined;
- (c) $\mathcal{J}$ is uniquely constructed from earlier source structure, in which case the construction and uniqueness theorem must be displayed.

Proof. Either the map is primitive in the model specification, allowed to vary over a class of models, or determined by the remaining structure. These cases are exhaustive. In the first, its physical content is assumed; in the second, there are distinct expansions; in the third, uniqueness is a theorem obligation. Naturality constrains a chosen $\mathcal{J}$ but does not by itself choose its codomain semantics or select one functor among all natural ones. $\square$

This theorem is essential for observer-first languages. The word “meaning” cannot make the source-to-physics map disappear. If laboratory meaning is already encoded in $\mathcal{J}$, it has entered at the axiom. If not, the physical lift remains to be derived.

9.9 T9: conservative-extension accounting

Theorem 9.9 (Added-premise accounting). *If*

$$T_h \not\models C \quad \text{but} \quad T_h \cup \Delta \models C,$$

then Δ is load bearing for C . The strongest accurate statement is that C follows from $T_h \cup \Delta$, not from T_h .

Proof. The first relation supplies a model of T_h in which C fails. Every proof of C from the extended theory therefore excludes at least that model through Δ . Deleting Δ restores the countermodel. $\square$

If Δ is an independently measured physical principle, the extended theory may be scientifically sound. But zero-input, three-axiom or pure source-recovery language must be withdrawn. If Δ merely says to select the observed target, it is circular.

9.10 T10: downstream non-determination

Theorem 9.10 (Downstream collapse schema). *Let F be any physical feature or downstream observable. If there exist P, Q with*

$$U(P) = U(Q) \quad \text{and} \quad F(P) \neq F(Q),$$

then T_{src} does not determine F .

Proof. Determination of F requires F to be constant on every realization fibre. The witness pair shows it is not constant on $\text{Real}(U(P))$. $\square$

This is how one countermodel can govern thousands of downstream lemmas. Any internal theorem whose conclusion is entirely source-algebraic may remain valid. Every physical conclusion that changes across the witness pair is not a consequence of the shared source theory.

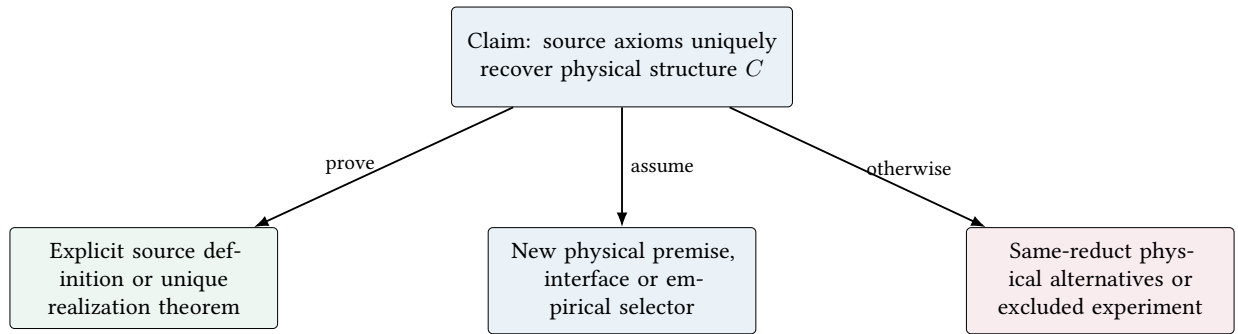
10 The master no-escape trilemma

No-escape theorem 10.1 (Physical realization trilemma). *For a version-locked source theory T_h and a claimed uniquely recovered physical structure C , at least one of the following is true:*

- I. Explicit derivation.** *A source-language definition or unique, functorial realization construction is supplied and proved to preserve all admitted physical readouts.*
- II. Additional physical structure.** *A realization map, physical selector, empirical attachment or equivalent premise is added beyond the neutral source theory.*
- III. Failure or restricted scope.** *Two observationally inequivalent physical expansions share the same source reduct, so unique physical recovery fails; alternatively the separating readout is excluded, in which case the theory does not recover that physical domain.*

No change of terminology, relabelling, gauge language, entropy maximization or later axiom addition produces a fourth possibility.

Proof. If the physical structure is uniquely determined in the first-order regime, [theorem 9.6](#) demands an explicit definition; in the general categorical regime, every realization fibre must be observationally singleton. This is case I. If the selector is instead present as primitive interpretation or an added interface, case II holds by [theorems 9.8](#) and [9.9](#). If neither is true, the physical relation is not fixed by the source; a same-reduct pair can differ on it, and any admitted separating readout invokes [theorem 9.3](#). If the readout is declared inadmissible, the theory has restricted its physical scope. Relabellings preserve reduct isomorphism; gauge transformations preserve readouts; source-factorized entropy cannot separate lifts by [theorem 9.7](#); later additions are counted by [theorem 9.9](#). The cases exhaust the logical possibilities. $\square$



There is no semantic fourth branch. “Implicit meaning” belongs to the first or second branch; “mere gauge” is blocked by observable separation; “maximum randomness” cannot leave the source fibre.

Figure 4: The exhaustive no-escape trilemma.

11 Application to the frozen OPH basis

11.1 What A1–A3 actually fix

The Mr Tickle reductio applies only to what the source language actually protects. It does not require changing A1’s abstract icosahedral packet. One keeps the same packet, algebras, records, overlaps, response and refinement maps and changes only the pictorial/presentational carrier. OPH’s own A1 reference explicitly treats embedding coordinates, materials and hidden presentation details as non-protected while retaining the abstract incidence and operational maps [2]. Therefore appearance cannot be the missing physical selector. If the source model is physically promoted, the promotion must come from something other than the fact that one drawing looks more scientific than another.

At the frozen commit, OPH’s A1 fixes a highly structured mathematical source: an oriented twelve-port carrier boundary, a federation and overlap nerve, an S^2 support tower, central record algebras, restriction and update

interfaces, and a faithful finite compact response algebra. A2 imposes naturality and internal implementation of accepted rechartings. A3 chooses an information projection on a fixed accessible algebra and fixed feasible family [2].

This is substantial mathematics. It is also not yet a unique physical world. The axiom reference itself states, in substance, that:

- A1 does not select laboratory identification, a particular global compact group, particles or couplings;
- A2 does not select a global compact group or laboratory interpretation;
- A3 selects a state inside one fixed model space and cannot compare unrelated field lists, Hilbert spaces or ontology classes without a common source space;
- later gravity, gauge, matter, flavour and quantitative conclusions must display the additional structure they consume;
- a named-realization witness proves existence inside that realization, not selection by the axioms.

These scope clauses are consistent with the theorem in this paper. They also mean that any public claim that A1–A3 alone uniquely recover the complete physical universe overstates the frozen formal basis.

11.2 The featureless source witness

The 2 August countermodel constructed an abstract oriented icosahedral carrier with twelve ports, a twelve-dimensional compact response algebra, internal A_5 action, functorial agreement and a maximum-randomness state. It recovered the conditional Lie-algebra type

$$\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$$

without giving any object a physical interpretation [1]. The result established the earlier branch-specific theorem:

$$\text{source algebraic recovery} \not\Rightarrow \text{physical Standard Model realization.}$$

OPH has since strengthened and expanded A1–A3. The present article therefore does not depend on every detail of the old finite witness satisfying every new clause. Instead it applies the universal fibre theorem to any source model that OPH itself accepts at the frozen commit. If no model satisfies the strengthened axioms, the source theory has no realization. If at least one model M does, the physical-lift question is governed by [theorem 10.1](#).

11.3 A bona fide physical pair: coupling freedom

To prevent the objection that one branch is “merely nonphysical”, consider two ordinary physical quantum field theories built over the same recovered local response algebra. Use the $\mathfrak{u}(1)$ factor and fix the same matter content, spacetime, masses, states and detector definitions. For each positive coupling e , define

$$\mathcal{L}_e = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu A_\mu\psi. \quad (5)$$

The source reduct remembers the abstract $\mathfrak{u}(1)$ response direction and all OPH source structure, but not the physical coupling unless a coupling selector has been added. Hence for $e \neq e'$,

$$U(P_e) = U(P_{e'}).$$

At tree level, a one-photon-exchange amplitude between charged matter states is proportional to e^2 , and its differential cross section is generically proportional to e^4 . At fixed non-singular kinematics,

$$\frac{d\sigma_e}{d\Omega} \neq \frac{d\sigma_{e'}}{d\Omega}.$$

The cross section is a calibrated, gauge-invariant physical record. Therefore

$$P_e \not\sim_O P_{e'}.$$

By [theorem 9.3](#), A1–A3 do not determine the physical coupling.

There are only two replies. OPH may add a unique coupling construction, in which case that construction is a new load-bearing bridge and must be audited. Or it may accept that the source algebra determines at most a response type, not the measured interaction strength.

11.4 A second physical pair: global gauge form

A local Lie algebra does not by itself fix the global gauge theory. Distinct global forms and line-operator spectra can define physically distinct four-dimensional gauge theories while sharing local algebraic data [15]. Thus even after recovering $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$, a global quotient, admissible line operators, matter representations and charge lattice remain separate selection obligations. If OPH derives these from additional carrier and matter premises, those premises belong in the target theorem. A local-algebra proof alone cannot carry the global physical conclusion.

11.5 A3 cannot choose between these lifts

The OPH A3 objective is evaluated on the fixed source state family. If P_e and $P_{e'}$ have the same source reduct, both the feasible source state and the source-relative entropy objective are equal. By [theorem 9.7](#), A3 cannot prefer e to e' . A reference state or constraint depending on e would solve a different problem in which the coupling has already entered the input.

11.6 The observer-language self-test

OPH makes records and public agreement central. That strengthens rather than weakens the no-go. If two physical lifts produce different stable public records, an observer-based theory cannot identify them without either deleting one admitted record or changing the source doctrine. If the theory excludes the scattering record in the coupling example, it does not recover the physical domain in which that scattering experiment is performed.

12 Exhaustive response classification

Attempted response	Formal requirement	Unavoidable consequence
“The countermodel is not physical.”	Give a source-derived predicate that accepts the intended physical model and rejects the witness.	The predicate is either a new physical premise or an explicit realization theorem.
“Physicality is supplied by an interface.”	Specify the interface and prove uniqueness/future-faithfulness.	If not source-definable, the interface is additional structure, not an A1–A3 theorem.
“A3 maximum randomness selects it.”	Show the feasible set or objective differs between the physical lifts without importing physical data.	If both factor through the source reduct they tie; if they do not, the selector contains extra physical information.
“They are gauge equivalent.”	Show every gauge-invariant readout agrees.	One differing cross section, spectrum, clock or detector count blocks the claim.
“That readout is inadmissible.”	Remove it from $\mathcal{O}$.	The theory no longer recovers the experimental domain containing that readout.
“The words were intended physically.”	Encode the intended semantics as a definition or axiom.	A definition must be proved unique; an axiom assumes the physical content.

Attempted response	Formal requirement	Unavoidable consequence
“The meaning is implicit.”	Establish implicit definability across all source models.	In first-order logic Beth yields an explicit definition; otherwise a same-reduct pair defeats uniqueness.
“We reinterpret the source objects.”	Supply an isomorphism or translation.	Isomorphic redescription preserves the counterexample; changing readouts or premises creates a new theory.
“We added a stronger axiom.”	Freeze the new theory at a new hash.	The added axiom is load bearing; the earlier theory remains non-entailing.
“We never claimed uniqueness.”	Restrict the claim to a conditional construction or named realization.	This accepts the paper’s conclusion and preserves only the narrower result.
“The Lean corpus proves the branch.”	Identify the exact theorem whose premises uniquely fix the physical lift.	Proof volume inside the source language cannot distinguish same-reduct physical expansions.
“The witness changes hidden implementation.”	Demonstrate that hidden change is future-inactive for all physical readouts.	If a physical readout changes, it is not hidden redundancy; if none changes, the witness is not observationally distinct.

No semantic escape

Every reply must modify the source premises, modify the observation family, or accept non-uniqueness. Renaming one of these moves does not change its logical status.

13 What the result refutes, preserves and requires

13.1 Preserved results

The following may remain correct:

- finite icosahedral incidence theorems;
- exact statements about a declared response representation;
- compact-Lie-algebra classifications under their hypotheses;
- naturality and overlap-consistency theorems;
- existence or exactness inside a named finite realization;
- information-projection theorems on a fixed feasible space;
- formal Lean proofs of those statements.

The paper does not attack a theorem merely because its words resemble physical nouns. It asks what the theorem’s formal premises entail.

13.2 Refuted overclaim

What fails is the unrestricted arrow

$$A1\text{--}A3 \text{ source structure} \implies \text{one uniquely determined physical world}$$

whenever the realization fibre contains observationally inequivalent lifts. The failure propagates to every downstream physical quantity that differs across those lifts by [theorem 9.10](#).

13.3 Requirements for a successful repair

A successful physical recovery theorem must supply all of the following:

1. a version-locked source theory;
2. a precisely typed physical target category;
3. a realization functor or relation;
4. existence for every relevant source model;
5. uniqueness up to observational equivalence;
6. preservation of gauge-invariant physical readouts;
7. proof that no target information was inserted into the source grammar, reference state, candidate menu or interpretation map;
8. independent empirical calibration where dimensions and couplings enter.

This is not an impossible standard. It is the exact content of the claim that a source architecture recovers physics rather than merely representing it.

14 Theorem-volume irrelevance: an exact quantitative result

The no-go can be sharpened into a quantitative statement about proof volume. Let $\tau_1, \dots, \tau_N$ be any finite family of propositions whose semantics factor through the frozen source reduct U . Define the joint source-theorem fingerprint

$$F_N(X) = (\mathbf{1}_{\tau_1(X)}, \dots, \mathbf{1}_{\tau_N(X)}) \in \{0, 1\}^N.$$

The family may contain theorems, checked lemmas, finite certificates, derived identities or any other Boolean judgements whose truth depends only on $U(X)$.

Theorem 14.1 (Theorem-fingerprint invariance). *If $U(P) = U(Q)$ and every τ_i factors through U , then for every finite N ,*

$$\boxed{F_N(P) = F_N(Q)}.$$

Consequently no deterministic classifier whose only input is the complete N -theorem fingerprint can distinguish P from Q .

Proof. For each i there exists a source predicate $\bar{\tau}_i$ such that $\tau_i = \bar{\tau}_i \circ U$. Hence

$$\tau_i(P) = \bar{\tau}_i(U(P)) = \bar{\tau}_i(U(Q)) = \tau_i(Q)$$

for every i . Equality holds coordinatewise, so the fingerprints are identical. Any post-processing map receives the same input on the two lifts and therefore returns the same output. $\square$

Define the physical discrimination count of the theorem family on a witness pair by

$$D_N(P, Q) := \sum_{i=1}^N \mathbf{1}[\tau_i(P) \neq \tau_i(Q)].$$

On a same-reduct pair, [theorem 14.1](#) gives

$$\boxed{D_N(P, Q) = 0} \quad \text{and, for } N > 0, \quad \boxed{\delta_N := D_N/N = 0}.$$

This is an exact zero, not a statement that the theorem family is merely “large” or “redundant”.

14.1 Information-theoretic form

The same result has a standard Shannon interpretation [16]. Let a binary random variable L choose equiprobably between two observationally inequivalent lifts P and Q with the same source reduct, and let

$$Y = F_N(L)$$

be the full theorem fingerprint observed by an analyst. Because $F_N(P) = F_N(Q)$, Y is constant with respect to L . Therefore

$$\boxed{I(L; Y) = 0 \text{ bits}}, \quad \boxed{H(L | Y) = H(L) = 1 \text{ bit}}.$$

Thus the entire source-theorem corpus supplies exactly zero information about which of the two physical lifts is realized. With k equiprobable, observationally inequivalent lifts in one source fibre,

$$I(L; Y) = 0, \quad H(L | Y) = \log_2 k.$$

The unresolved physical information is independent of N .

What “zero” means

This result does *not* say that the source theorems are mathematically meaningless. It says something narrower and exactly measurable: their joint truth vector has zero discriminatory power for choosing among physically inequivalent lifts that share the same source reduct. A theorem can be perfectly correct internally while contributing zero bits to the missing source-to-physics selection problem.

14.2 Numerical instantiation for OPH

OPH’s public 25 August 2026 commit states that its automated “theorem floors” were raised to 8400 [17]; the live README checked on 26 August advertises more than 8800 Lean theorems [3]. Running OPH’s own `tools/check_lean_theorem_count.py` against the frozen checkout used for this audit returns 8449 public theorem/lemma declarations across 381 Lean source files; the exact output and counting-script hash accompany this article in `OPH_THEOREM_COUNT_AUDIT.txt`.

The count itself does not establish that all 8449 declarations are source-internal. Instead the corpus is partitioned by a mathematically sharp criterion. Let

$$S = \{i : \tau_i \text{ factors through } U\}, \quad E = \{1, \dots, 8449\} \setminus S.$$

Every declaration in S is blind on the same-reduct witness pair, so

$$D_S(P, Q) = 0, \quad I(L; F_S) = 0 \text{ bits}.$$

This includes any 5000-, 8400-, or 8449-element family for which source factorization is actually established.

Conversely, if a declaration τ_j does distinguish the two physical lifts,

$$\tau_j(P) \neq \tau_j(Q),$$

then [theorem 14.1](#) immediately implies that τ_j cannot factor through the common source reduct. Its discriminating content therefore comes from structure outside that reduct: an additional physical premise, realization map, calibration, target attachment, or strengthened axiom. It may be a valid theorem, but it cannot simultaneously be cited as evidence that the original common source structure alone selected the physical lift.

Thus the raw number 8449 has no standalone evidentiary force for unique physical recovery. Each declaration falls into one of two relevant bins:

$$\boxed{\text{source-factorized} \Rightarrow 0 \text{ lift-discrimination bits}}$$

or

$$\boxed{\text{lift-separating} \Rightarrow \text{beyond-source structure is load bearing.}}$$

There is no third category in which a theorem both depends only on the common source reduct and nevertheless

distinguishes two models that have that same reduct.

Theorem 14.2 (Theorem volume cannot repair physical non-definability). *Let P, Q share a source reduct and be separated by an admitted physical readout. Adjoining any finite family of source-language theorems leaves both facts unchanged:*

$$F_N(P) = F_N(Q) \quad \text{and} \quad P \not\sim_{\text{obs}} Q.$$

Hence no finite increase in source-internal theorem count can convert the non-singleton realization fibre into a unique physical recovery.

This result isolates the only scientifically relevant escape. A new result must make a physical distinction that the old source reduct did not make. Then it is either (i) an explicit derivation of additional physical structure from stronger premises, or (ii) an additional physical premise/identification. It cannot be credited retroactively to theorem volume already blind to that distinction.

15 Architectural convergence as a role-graph problem

The conceptual-ancestry question cannot be evaluated by counting shared words. The relevant object is a dependency architecture. Let

$$G_P = (V_P, E_P, w)$$

be the role graph extracted from a dated prior corpus, where vertices are functional obligations—realization gate, record closure, dynamics before action, admissible history before minimization, countermodel uniqueness, premise ancestry, no-privilege selection, observer-history retention and physical calibration—and edges record their logical order. Let $x_t(v) \in [0, 1]$ be the auditor-coded visibility of role v in a version-locked OPH snapshot at time t .

For role weights w_v and probative factors p_v , define

$$C_V(t) = \frac{\sum_v w_v p_v x_t(v)}{\sum_v w_v p_v}.$$

For a directed edge $e = (u, v)$, define

$$x_t(e) = \min\{x_t(u), x_t(v)\}, \quad C_E(t) = \frac{\sum_e w_e p_e x_t(e)}{\sum_e w_e p_e}, \quad p_e = \min\{p_u, p_v\}.$$

These quantities are not probabilities of copying. They are transparent arithmetic conditional on a published coding table. Every role, factor, weight, snapshot value and evidence group is supplied in machine-readable form so that a critic can change one entry and recompute the result.

15.1 Frozen coding, robustness and negative controls

The coding down-weights or excludes themes for which OPH has a documented earlier public antecedent: generic Born-rule work, Lorentz/3D kinematics, broad Yukawa/mixing interest and ordinary A_5 /golden-ratio representation theory. High-weight roles are unusual ordering and governance functions: source/physical separation, dynamics to action class, admissible history to information minimum to physical action, record/readout closure, alternative-realization countermodels, premise ancestry, no-privilege promotion, observer index versus accumulated proper history, calibration and explicit open-row custody.

Every coded role is non-decreasing across the frozen snapshots. Consequently, for any non-negative alternative weights a_v ,

$$\tilde{C}(t) = \frac{\sum_v a_v x_t(v)}{\sum_v a_v}$$

is non-decreasing. Thus reweighting changes magnitude but cannot reverse the direction of the coded trend. The repository also supplies unweighted counts above four visibility thresholds.

Under the frozen coding, probative role alignment rises from 0.126 at the 1 August pre-audit public baseline to 0.955 at the V3.34 snapshot. The probative edge score rises from 0.090 to 0.928. The largest changes occur in the 5–7 August mechanics wave, the 12–17 August governance wave and the 25–26 August clock/force/calibration waves, not in excluded Born/Lorentz controls.

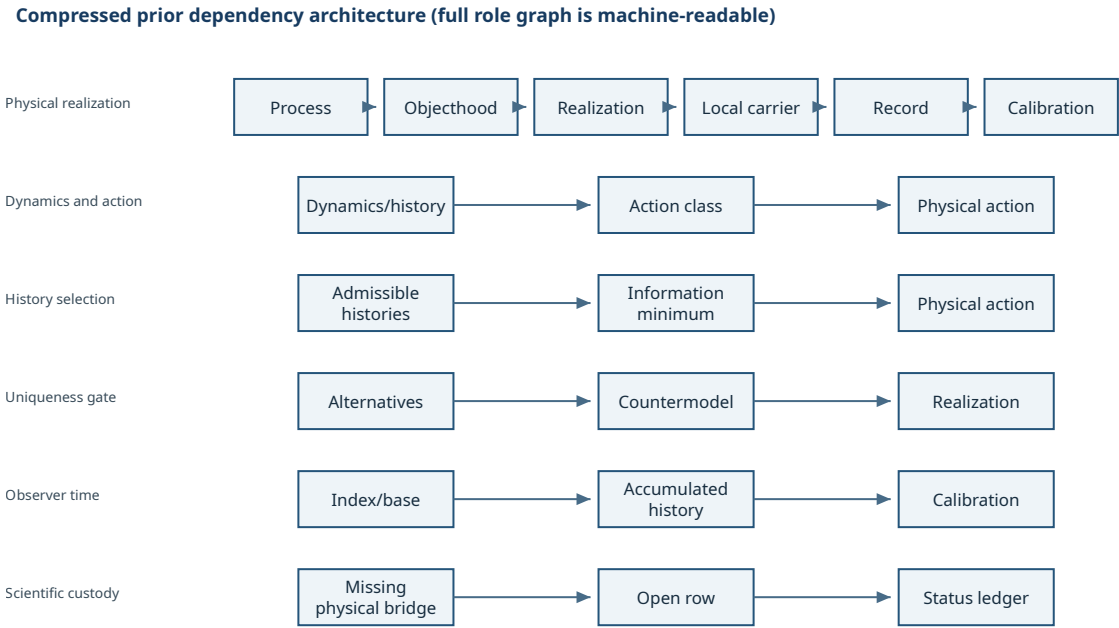


Figure 5: Compressed view of the prior functional dependency architecture. The complete weighted role graph is distributed as JSON/DOT/SVG. Nodes are scientific functions rather than vocabulary; generic Born, Lorentz and ordinary icosahedral/golden structures are controls.

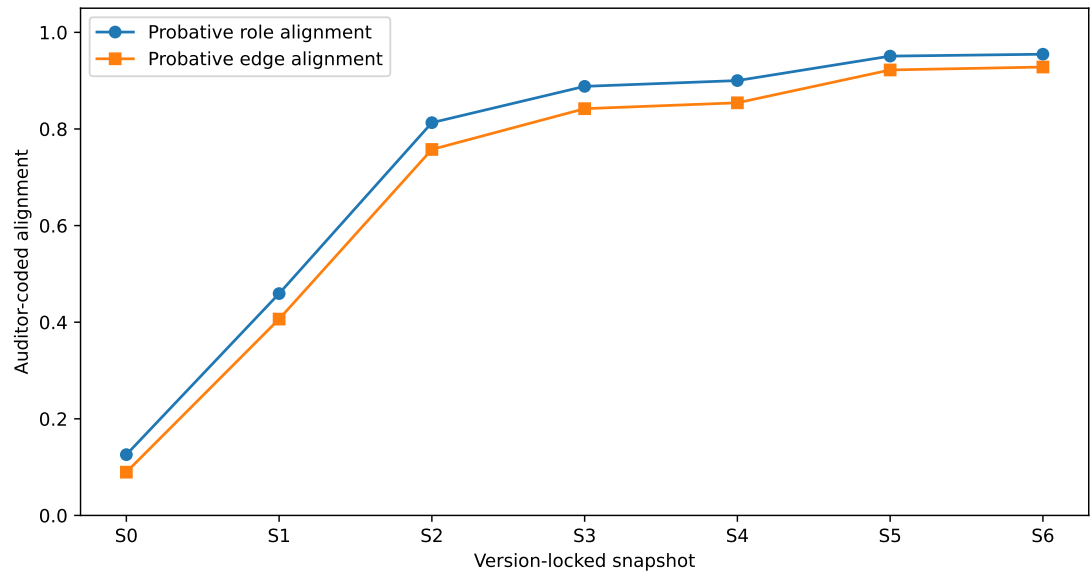


Figure 6: Auditor-coded structural alignment across version-locked OPH snapshots. The curves exclude low-probative controls. They are not plagiarism probabilities; they quantify role and dependency-edge coverage under the frozen public coding.

15.2 The 25–26 August sequence

Commit 4bc014ab lands recurrence clock, charge-fixed interaction, internal-energy inertia and local energy balance while saying that the rows remain open and nothing is discharged. Commit fe4c9579 rebuilds the flagship around a three-axiom compression thesis while naming five standing unsupplied maps. Commit 864051be lands force law, speed limit, index-versus-proper-length clock rate and a golden split, with declared identifications, alternatives, open physical-attachment rows and explicit falsifiers. Its child 05207466 adds a proper-length-clocked chain, in-block force law, golden-sector irreducibility and carrier flow, again declaring units and clocking rules while discharging no physical row.

The inference licensed by this chronology is architectural convergence, not subjective intent. The appropriate next step is a dated conceptual-ancestry statement: either exhibit a pre-contact trail for the selected ordering or disclose material external influence, including AI context, and attribute it.

16 Conceptual ancestry: recipe versus ingredients

Independent code, local lemmas and implementation certificates establish implementation authorship. They do not by themselves establish independent selection of the conceptual recipe: which problems are load-bearing, the order in which they are crossed, the role each bridge plays, which alternatives must be retained and what closes a physical claim. We therefore distinguish

independent manufacture from independent conceptual ancestry.

The public evidence ledger records directly provable dates and hashes, role-preserving structural inferences, contextual convergences and explicit negative controls. It does not convert structural similarity into a legal or intentional finding.

The requested disclosure concerns ten functions: dynamics before action; admissible history before minimization; source/physical realization gates; countermodels as uniqueness tests; premise ancestry and claim-status custody; no-privilege selection; record/readout closure; observer index versus accumulated proper history; force/current/Hamiltonian/calibration gaps; and programme compression into a small dependency spine. For each, a dated pre-contact OPH trail or disclosure of external/AI-context influence would resolve the public ancestry question far more directly than another local proof certificate.

17 Lean 4 proof kernel

The accompanying file `PhysicalNonDefinability.lean` formalizes the abstract logical kernel without importing OPH code. This separation is intentional. The universal theorem should not depend on OPH's implementation choices. In addition to observational equivalence, source determination, unique realization fibres, factorization and theorem-volume invariance, the file now formalizes the reductio itself. Named results include:

- `no_semantic_alchemy`;
- `no_source_only_enrichment_spawns_physics`;
- `physical_distinction_blocks_public_factorization`;
- `same_screen_distinct_boundary_theories_refute_screen_selector`;
- `same_screen_different_bulk_refutes_bare_screen_holography`;
- the concrete `mr_tickle_screen_reductio`.

The source also contains count-independent theorem-family blindness and explicit 5000, 8400, 8449 and 8800 convenience corollaries.

The shortest master result is structurally:

```
theorem no_escape_kernel
  (red : Phys -> Src) (readout : Obs -> Phys -> Val)
  {p q : Phys} {o : Obs}
  (hred : red p = red q)
  (hsep : readout o p != readout o q) :
  not (SourceDetermines red (ObsEq readout))
```

An ASCII rendering appears in [section B](#); the exact UTF-8 Lean source accompanies the article. The present build environment did not contain a Lean executable, so the article does not represent the supplement as compiled here. Its statements use only elementary Lean 4 definitions and are deliberately small enough for independent checking.

18 Discussion

18.1 Why one metatheorem outweighs a large proof surface

A proof assistant checks whether conclusions follow from encoded premises. It does not decide whether the premises are physically necessary, whether a formal object has the claimed laboratory meaning, or whether an unencoded realization map is unique. Thus a large theorem count can coexist with a single missing physical bridge.

The appropriate comparison is not

number of lemmas versus number of lemmas,

but

source theory $\longrightarrow$ realization fibre $\longrightarrow$ physical readouts.

If the middle fibre is non-singleton, every source-internal proof leaves the physical ambiguity untouched.

18.2 Why the theorem is not anti-formalization

Formalization is especially valuable here. It prevents a theory from moving silently between source structure and physical interpretation. The no-go suite should be imported alongside OPH's own modules and instantiated with its exact source-reduct function, physical-realization type and readout family. A successful OPH reply would be a formal uniqueness theorem. A failed instantiation would identify the exact additional premise required.

18.3 Relation to underdetermination in physics

Physical theories routinely share local algebraic data while differing in global structure, couplings, matter content, topological sectors or line operators. The global-form example in gauge theory is not a philosophical possibility but standard field-theoretic structure [15]. The no-go theorem therefore does not impose an artificial standard. It asks a reconstruction programme to distinguish alternatives that established physics already distinguishes.

18.4 Strongest defensible status for OPH after the reductio

At the frozen commit, OPH's own seven-value status discipline can absorb the result cleanly:

- source-internal universal results may be `axiom_forced`;
- exhaustively checked finite constructions may be `exact_named_realization`;
- unclosed lifts remain `conditional_open_interface`;
- countermodel-limited statements are `independence_limited`;
- laboratory mappings are `physical_identification`.

The no-go demands that these distinctions govern headline claims as well as registries. If they do, OPH is a conditional reconstruction programme with a large formal apparatus. If they do not, the broader physical claim is not entailed by the frozen axioms.

19 Conclusion

The Mr Tickle Screen makes the foundational issue difficult to disguise. A formal source structure can be represented on a scientific-looking sphere, on a long-armed orange cartoon, or in a text file. If those presentations carry the same protected source structure, source theorems cannot distinguish them. Nothing about one presentation causes

nature to instantiate its referent. That missing relation is external to source syntax unless physical semantics has been imported as additional structure.

The result is exact. *No Semantic Alchemy* proves that no satisfiable source-only enrichment can force an unconstrained external realization relation to be inhabited. *No boundary theory from bare screen* proves that one finite carrier can support inequivalent boundary dynamics. *No Bulk From Bare Boundary* proves that a bulk assignment cannot factor through a bare screen when those boundary theories reconstruct differently. HKLL and related holographic machinery do not rescue the bare screen: they begin with a specified boundary quantum theory and a reconstruction dictionary, precisely the objects that bare incidence and observer consensus do not supply.

For OPH, the strongest rigorous conclusion is therefore stronger than “some physical bridges remain open.” As long as the programme remains a source-only derivation from A1–A3 and their formal consequences, physical closure is impossible. A future successful physical closure would have to supply independently grounded boundary dynamics, physical realization, calibration and a valid holographic dictionary. Those additions could make a new theory scientifically substantive, but they would also identify the load-bearing physical structure that the original three-axiom screen did not contain.

Thus the final no-escape statement is

formal screen $\nrightarrow$ physical boundary theory $\nrightarrow$ holographic dictionary $\nrightarrow$ physical bulk.

The crossed arrows are not invitations to add more decorative source formalism. Each is a separate existence/uniqueness/realization obligation. Looking harder at the drawing does not put a universe behind it.

Methods

Repository audit. The OPH basis was frozen to the commit stated in [section 6](#). The reader-facing axiom reference and README were treated as primary public sources. No inference in the theorem suite depends on mutable branch-head language after that hash.

Formal method. The proof separates a source signature from a physical extension signature, forms the reduct functor, and defines physical uniqueness as observational singleton fibres. Standard reduct invariance supplies the core indiscernibility result. Beth definability is used only as an additional first-order closure; the direct two-expansion proof applies more generally.

Physical witness discipline. A counterexample counts only when the models share the frozen source reduct and differ in an admitted gauge-invariant readout. Differences of notation, coordinates, labels or hidden implementation do not qualify.

Holographic typing audit. The holography section uses Maldacena/Witten/HKLL/ADH only for the established typing of the reconstruction problem: a specified boundary quantum theory and duality/reconstruction data precede the reconstruction of bulk operators. No claim is made that HKLL itself proves ontological reality. The audited claim is narrower: a bare finite screen is not, by itself, the boundary quantum theory that HKLL consumes.

Live-head separation. The theorem is proved against a frozen baseline so later changes cannot be projected backward. A separate 26 August public-head snapshot is cited only to show that the current programme still advertises an emergent-reality/ holographic-screen reconstruction and a theorem count exceeding 8800.

Data and code availability

The primary manuscript source and abstract Lean 4 theorem kernel accompany this article. The audited OPH source is publicly versioned by the frozen Git commit. The author’s 2 August 2026 featureless countermodel is preserved as both LaTeX and PDF with hashes recorded in the associated provenance dossier.

Author contributions

The public package is issued anonymously. The technical audit, role coding, source custody and formal kernel are open to public correction through the repository workflow.

Competing interests

The author declares no financial competing interests. This article is an adversarial scientific audit of a public theoretical programme and is not a legal adjudication of authorship or intent.

A Formal response checklist

For each disputed OPH claim C , a reproducible audit should publish:

1. frozen commit and file hashes;
2. complete dependency list for C ;
3. source signature L_{src} ;
4. physical extension signature L_{phys} ;
5. reduct map U ;
6. physical realization relation or functor;
7. admitted physical readout family $\mathcal{O}$;
8. proof of existence of a physical lift;
9. proof of observational uniqueness of the lift;
10. countermodel search over alternative lifts;
11. explicit classification of every added premise;
12. statement of the strongest surviving claim if uniqueness fails.

A response that omits items 3–9 has not answered the theorem. It has changed vocabulary without closing the realization fibre.

B Lean 4 theorem kernel

```
import Init

/--!
PhysicalNonDefinability.lean

Abstract Lean 4 kernel for the no-escape theorem suite accompanying:
"The Mr Tickle Screen: a reductio ad absurdum against physical closure from bare observer screens".

The file deliberately uses only elementary logical structure. It does not
import OPH modules and does not assert that any particular OPH structure
exists. Instead it proves the universal metatheorems used in the paper:
same source reduct + different physical readout defeats source determination.
-/

universe uSrc uPhys uObs uVal uCost uFeature

namespace PhysicalNonDefinability

/-- Two physical models are observationally equivalent when every admitted
readout returns the same value. -/
def ObsEq {Phys : Type uPhys} {Obs : Type uObs} {Val : Type uVal}
  (readout : Obs -> Phys -> Val) (p q : Phys) : Prop :=
  forall o, readout o p = readout o q

/-- A source reduct determines physical models up to the supplied equivalence. -/
def SourceDetermines {Src : Type uSrc} {Phys : Type uPhys}
  (red : Phys -> Src) (eqv : Phys -> Phys -> Prop) : Prop :=
  forall p q, red p = red q -> eqv p q

/-- A realization relation has a unique physical fibre up to equivalence. -/
def UniqueFibre {Src : Type uSrc} {Phys : Type uPhys}
  (realizes : Src -> Phys -> Prop) (eqv : Phys -> Phys -> Prop) : Prop :=
  forall s p q, realizes s p -> realizes s q -> eqv p q

/-- A physical-valued function factors through the source reduct. -/
def FactorsThrough {Src : Type uSrc} {Phys : Type uPhys} {A : Type _}
  (red : Phys -> Src) (f : Phys -> A) : Prop :=
  exists f0 : Src -> A, forall p, f p = f0 (red p)

/-- A predicate factors through the source reduct. -/
def PredicateFactorsThrough {Src : Type uSrc} {Phys : Type uPhys}
  (red : Phys -> Src) (K : Phys -> Prop) : Prop :=
  exists K0 : Src -> Prop, forall p, K p <=> K0 (red p)

/-- A minimizer of an objective over an admissible physical class. -/
def Minimizer {Phys : Type uPhys} {Cost : Type uCost} [Preorder Cost]
```

```

    (admissible : Phys -> Prop) (objective : Phys -> Cost) (p : Phys) : Prop :=
    admissible p /\ forall q, admissible q -> objective p <= objective q

/-- T1: source-language predicates cannot distinguish equal reducts. -/
theorem source_predicate_indiscernibility
  {Src : Type uSrc} {Phys : Type uPhys}
  (red : Phys -> Src) (P : Src -> Prop) {p q : Phys}
  (hred : red p = red q) :
  P (red p) <=> P (red q) := by
  constructor
  * intro hp
    rw [hred] at hp
    exact hp
  * intro hq
    rw [← hred] at hq
    exact hq

/-- T2: one same-reduct inequivalent pair refutes source determination. -/
theorem witness_refutes_source_determination
  {Src : Type uSrc} {Phys : Type uPhys}
  (red : Phys -> Src) (eqv : Phys -> Phys -> Prop)
  {p q : Phys} (hred : red p = red q) (hneqv : ! eqv p q) :
  ! SourceDetermines red eqv := by
  intro hdet
  exact hneqv (hdet p q hred)

/-- T3: two inequivalent realizations of one source refute unique recovery. -/
theorem two_realizations_refute_unique_fibre
  {Src : Type uSrc} {Phys : Type uPhys}
  (realizes : Src -> Phys -> Prop) (eqv : Phys -> Phys -> Prop)
  {s : Src} {p q : Phys}
  (hp : realizes s p) (hq : realizes s q) (hneqv : ! eqv p q) :
  ! UniqueFibre realizes eqv := by
  intro huniq
  exact hneqv (huniq s p q hp hq)

/-- T4: an admitted observable difference defeats observational equivalence. -/
theorem observable_witness_refutes_obsEq
  {Phys : Type uPhys} {Obs : Type uObs} {Val : Type uVal}
  (readout : Obs -> Phys -> Val) {p q : Phys} {o : Obs}
  (hsep : readout o p != readout o q) :
  ! ObsEq readout p q := by
  intro h
  exact hsep (h o)

/-- T5: gauge equivalence cannot relate models separated by a gauge-invariant readout. -/
theorem observable_separation_blocks_gauge_escape
  {Phys : Type uPhys} {Obs : Type uObs} {Val : Type uVal}
  (readout : Obs -> Phys -> Val) (gaugeEq : Phys -> Phys -> Prop)
  (gauge_preserves : forall {p q}, gaugeEq p q -> ObsEq readout p q)
  {p q : Phys} (hsep : ! ObsEq readout p q) :
  ! gaugeEq p q := by
  intro hg
  exact hsep (gauge_preserves hg)

/-- T6: a source-factorized quantity is equal on equal reducts. -/
theorem factorized_quantity_equal_on_same_reduct
  {Src : Type uSrc} {Phys : Type uPhys} {A : Type _}
  (red : Phys -> Src) (f : Phys -> A)
  (hfactor : FactorsThrough red f)
  {p q : Phys} (hred : red p = red q) :
  f p = f q := by
  cases hfactor with
  | intro f0 hf0 =>
    calc
      f p = f0 (red p) := hf0 p
      _ = f0 (red q) := congrArg f0 hred
      _ = f q := (hf0 q).symm

/-- T7: a source-factorized admissibility predicate is equal on equal reducts. -/
theorem factorized_predicate_equal_on_same_reduct
  {Src : Type uSrc} {Phys : Type uPhys}
  (red : Phys -> Src) (K : Phys -> Prop)
  (hfactor : PredicateFactorsThrough red K)
  {p q : Phys} (hred : red p = red q) :
  K p <=> K q := by
  cases hfactor with
  | intro K0 hK0 =>
    constructor
    * intro hp
      have hp0 : K0 (red p) := (hK0 p).mp hp
      have hq0 : K0 (red q) := by
        rw [← hred]
        exact hp0
      exact (hK0 q).mpr hq0
    * intro hq
      have hq0 : K0 (red q) := (hK0 q).mp hq

```

```

    have hp0 : K0 (red p) := by
      rw [hred]
      exact hq0
    exact (hK0 p).mpr hp0

/-- T8: an information criterion and feasible set that both factor through
source data cannot select uniquely between distinct same-reduct lifts. -/
theorem source_blind_optimization_cannot_select_unique_lift
  {Src : Type uSrc} {Phys : Type uPhys} {Cost : Type uCost}
  [Preorder Cost]
  (red : Phys -> Src) (admissible : Phys -> Prop) (objective : Phys -> Cost)
  (hK : PredicateFactorsThrough red admissible)
  (hJ : FactorsThrough red objective)
  {p q : Phys} (hred : red p = red q)
  (hp : Minimizer admissible objective p) (hpq : p != q) :
  ! (forall r, Minimizer admissible objective r -> r = p) := by
  have hKpq : admissible p <=> admissible q :=
    factorized_predicate_equal_on_same_reduct red admissible hK hred
  have hJpq : objective p = objective q :=
    factorized_quantity_equal_on_same_reduct red objective hJ hred
  have hq : Minimizer admissible objective q := by
    constructor
    * exact hKpq.mpr hp.1
    * intro r hr
      calc
        objective q = objective p := hJpq.symm
        _ <= objective r := hp.2 r hr
  intro hunique
  have hqp : q = p := hunique q hq
  exact hqp.symm

/-- T9: any downstream feature differing on one same-reduct pair is not
source-determined. -/
def DeterminesFeature {Src : Type uSrc} {Phys : Type uPhys}
  {Feature : Type uFeature}
  (red : Phys -> Src) (feature : Phys -> Feature) : Prop :=
  forall p q, red p = red q -> feature p = feature q

theorem feature_witness_refutes_determination
  {Src : Type uSrc} {Phys : Type uPhys} {Feature : Type uFeature}
  (red : Phys -> Src) (feature : Phys -> Feature)
  {p q : Phys} (hred : red p = red q)
  (hdiff : feature p != feature q) :
  ! DeterminesFeature red feature := by
  intro hdet
  exact hdiff (hdet p q hred)

/-- T10: composing with a downstream readout does not restore determination
when that readout separates a same-reduct pair. -/
theorem downstream_readout_difference_refutes_determination
  {Src : Type uSrc} {Phys : Type uPhys}
  {Feature : Type uFeature} {Val : Type uVal}
  (red : Phys -> Src) (feature : Phys -> Feature) (readout : Feature -> Val)
  {p q : Phys} (hred : red p = red q)
  (hdiff : readout (feature p) != readout (feature q)) :
  ! DeterminesFeature red (fun x => readout (feature x)) := by
  exact feature_witness_refutes_determination red
    (fun x => readout (feature x)) hred hdiff

/-- Master kernel: same source reduct plus one admitted physical difference
refutes the claim that source structure determines physical reality. -/
theorem no_escape_kernel
  {Src : Type uSrc} {Phys : Type uPhys} {Obs : Type uObs} {Val : Type uVal}
  (red : Phys -> Src) (readout : Obs -> Phys -> Val)
  {p q : Phys} {o : Obs}
  (hred : red p = red q) (hsep : readout o p != readout o q) :
  ! SourceDetermines red (ObsEq readout) := by
  apply witness_refutes_source_determination red (ObsEq readout) hred
  exact observable_witness_refutes_obsEq readout hsep

end PhysicalNonDefinability

namespace PhysicalNonDefinability

/-! ## Theorem-volume irrelevance on a nontrivial realization fibre -/

universe uLabel

/-- A finite family of Boolean-valued source-language judgements factors
through the source reduct when every judgement depends only on source data. -/
def TheoremFamilyFactorsThrough {Src : Type uSrc} {Phys : Type uPhys}
  (red : Phys -> Src) {n : Nat} (judge : Fin n -> Phys -> Bool) : Prop :=
  exists judge0 : Fin n -> Src -> Bool,
    forall i p, judge i p = judge0 i (red p)

/-- The joint truth-value fingerprint of a finite theorem/judgement family. -/
def TheoremFingerprint {Phys : Type uPhys} {n : Nat}

```

```

(judge : Fin n -> Phys -> Bool) (p : Phys) : Fin n -> Bool :=
fun i => judge i p

/-- A theorem family distinguishes two physical lifts when at least one
source-language judgement has a different truth value on the two lifts. -/
def FamilyDistinguishes {Phys : Type uPhys} {n : Nat}
  (judge : Fin n -> Phys -> Bool) (p q : Phys) : Prop :=
exists i, judge i p != judge i q

/-- T11: every finite source-language theorem fingerprint is identical on
physical lifts with the same source reduct. The result is independent of n. -/
theorem theorem_fingerprint_equal_on_same_reduct
  {Src : Type uSrc} {Phys : Type uPhys} {n : Nat}
  (red : Phys -> Src) (judge : Fin n -> Phys -> Bool)
  (hfactor : TheoremFamilyFactorsThrough red judge)
  {p q : Phys} (hred : red p = red q) :
  TheoremFingerprint judge p = TheoremFingerprint judge q := by
cases hfactor with
| intro judge0 hjudge =>
  funext i
  calc
    judge i p = judge0 i (red p) := hjudge i p
    _ = judge0 i (red q) := congrArg (judge0 i) hred
    _ = judge i q := (hjudge i q).symm

/-- T12: no finite number of source-language theorems can distinguish two
same-reduct physical lifts. -/
theorem source_theorem_family_zero_discrimination
  {Src : Type uSrc} {Phys : Type uPhys} {n : Nat}
  (red : Phys -> Src) (judge : Fin n -> Phys -> Bool)
  (hfactor : TheoremFamilyFactorsThrough red judge)
  {p q : Phys} (hred : red p = red q) :
  ! FamilyDistinguishes judge p q := by
intro hdist
cases hdist with
| intro i hi =>
  have hfp : TheoremFingerprint judge p = TheoremFingerprint judge q :=
    theorem_fingerprint_equal_on_same_reduct red judge hfactor hred
  exact hi (congrFun hfp i)

/-- T13: post-processing the complete source-theorem fingerprint cannot
restore information that is absent from the fingerprint itself. -/
theorem any_classifier_on_source_fingerprint_is_blind
  {Src : Type uSrc} {Phys : Type uPhys} {Label : Type uLabel} {n : Nat}
  (red : Phys -> Src) (judge : Fin n -> Phys -> Bool)
  (hfactor : TheoremFamilyFactorsThrough red judge)
  (classify : (Fin n -> Bool) -> Label)
  {p q : Phys} (hred : red p = red q) :
  classify (TheoremFingerprint judge p) =
  classify (TheoremFingerprint judge q) := by
exact congrArg classify
  (theorem_fingerprint_equal_on_same_reduct red judge hfactor hred)

/-- T14: theorem volume cannot repair physical non-definability. Even after
adjoining an arbitrary finite source-language theorem family, a physical
readout that separates two same-reduct lifts still refutes source determination. -/
theorem theorem_volume_cannot_repair_physical_nondefinability
  {Src : Type uSrc} {Phys : Type uPhys} {Obs : Type uObs} {Val : Type uVal}
  {n : Nat}
  (red : Phys -> Src) (readout : Obs -> Phys -> Val)
  (judge : Fin n -> Phys -> Bool)
  (hfactor : TheoremFamilyFactorsThrough red judge)
  {p q : Phys} {o : Obs}
  (hred : red p = red q) (hsep : readout o p != readout o q) :
  (! FamilyDistinguishes judge p q) /\
  (! SourceDetermines red (ObsEq readout)) := by
constructor
* exact source_theorem_family_zero_discrimination red judge hfactor hred
* exact no_escape_kernel red readout hred hsep

/-- Numerical corollary at the user's requested scale: 5,000 source-language
judgements still have zero discrimination on a same-reduct witness pair. -/
theorem five_thousand_source_theorems_zero_discrimination
  {Src : Type uSrc} {Phys : Type uPhys}
  (red : Phys -> Src) (judge : Fin 5000 -> Phys -> Bool)
  (hfactor : TheoremFamilyFactorsThrough red judge)
  {p q : Phys} (hred : red p = red q) :
  ! FamilyDistinguishes judge p q := by
exact source_theorem_family_zero_discrimination red judge hfactor hred

/-- Public OPH theorem-floor instantiation from 25 August 2026: even 8,400
source-language judgements cannot distinguish a same-reduct witness pair. -/
theorem eight_thousand_four_hundred_source_theorems_zero_discrimination
  {Src : Type uSrc} {Phys : Type uPhys}
  (red : Phys -> Src) (judge : Fin 8400 -> Phys -> Bool)
  (hfactor : TheoremFamilyFactorsThrough red judge)
  {p q : Phys} (hred : red p = red q) :

```

```

! FamilyDistinguishes judge p q := by
exact source_theorem_family_zero_discrimination red judge hfactor hred

/-- The result is count-independent; 8,449 is included as a convenience for
checkouts reporting that declaration count. -/
theorem eight_thousand_four_hundred_forty_nine_source_theorems_zero_discrimination
{Src : Type uSrc} {Phys : Type uPhys}
(red : Phys -> Src) (judge : Fin 8449 -> Phys -> Bool)
(hfactor : TheoremFamilyFactorsThrough red judge)
{p q : Phys} (hred : red p = red q) :
! FamilyDistinguishes judge p q := by
exact source_theorem_family_zero_discrimination red judge hfactor hred

/-- T15: if one Boolean judgement actually separates two same-reduct lifts,
that judgement cannot factor solely through the source reduct. This is the
formal source/extra-structure dichotomy for any allegedly load-bearing proof. -/
theorem separating_judgement_is_not_source_factorized
{Src : Type uSrc} {Phys : Type uPhys}
(red : Phys -> Src) (judge : Phys -> Bool)
{p q : Phys} (hred : red p = red q)
(hsep : judge p != judge q) :
! FactorsThrough red judge := by
intro hfactor
exact hsep (factorized_quantity_equal_on_same_reduct red judge hfactor hred)

end PhysicalNonDefinability

namespace PhysicalNonDefinability

/-! ## No Semantic Alchemy and no bulk from a bare boundary -/

universe uScreen uBoundary uBulk uWorld uIndex

/-- A source predicate would "force external realization" only if every
source model satisfying it had a realization for every possible external
realization relation. This deliberately separates source syntax/semantics
from an external physical-instantiation relation. -/
def ForcesExternalRealization {Src : Type uSrc} {World : Type uWorld}
(T : Src -> Prop) : Prop :=
forall s, T s -> forall realizes : Src -> World -> Prop, exists w, realizes s w

/-- T16 (No Semantic Alchemy): a satisfiable source theory cannot, solely by
its source-internal content, force an unconstrained external realization
relation to be inhabited. Choosing the empty relation is the countermodel. -/
theorem no_semantic_alchemy
{Src : Type uSrc} {World : Type uWorld}
(T : Src -> Prop) (hSat : exists s, T s) :
! ForcesExternalRealization (World := World) T := by
intro hforce
obtain <s, hs> := hSat
obtain <w, hw> := hforce s hs (fun _ _ => False)
exact hw

/-- A family of source-only constraints. -/
def SourceConstraintFamily {I : Type uIndex} {Src : Type uSrc}
(A : I -> Src -> Prop) (s : Src) : Prop :=
forall i, A i s

/-- T17 (No source-only enrichment): arbitrarily many additional constraints
that remain predicates of the source object do not create an external physical
realization. If the enriched source theory is satisfiable, the empty external
realization relation remains an admissible expansion. -/
theorem no_source_only_enrichment_spawns_physics
{I : Type uIndex} {Src : Type uSrc} {World : Type uWorld}
(A : I -> Src -> Prop)
(hSat : exists s, SourceConstraintFamily A s) :
! ForcesExternalRealization (World := World) (SourceConstraintFamily A) := by
exact no_semantic_alchemy (World := World) (SourceConstraintFamily A) hSat

/-- T18 (public-world factorization obstruction): if a purported public-world
quotient identifies two physical states that an admitted physical observable
separates, that observable cannot factor through the public quotient. -/
theorem physical_distinction_blocks_public_factorization
{Phys : Type uPhys} {Public : Type uScreen} {Out : Type uVal}
(publicView : Phys -> Public) (observable : Phys -> Out)
{p q : Phys}
(hpublic : publicView p = publicView q)
(hphysical : observable p != observable q) :
! FactorsThrough publicView observable := by
intro hfactor
exact hphysical
(factorized_quantity_equal_on_same_reduct
publicView observable hfactor hpublic)

/-- A bare boundary/screen is sufficient for a boundary theory only if one
selector reconstructs every boundary theory from its bare screen. -/
def BareScreenSelectsBoundary

```

```

    {Screen : Type uScreen} {Boundary : Type uBoundary}
    (screenOf : Boundary -> Screen) : Prop :=
    exists select : Screen -> Boundary, forall b, select (screenOf b) = b

/-- T19 (No boundary theory from bare screen): if two distinct boundary
quantum theories have the same bare screen, no screen-only selector can recover
the boundary theory. -/
theorem same_screen_distinct_boundary_theories_refute_screen_selector
  {Screen : Type uScreen} {Boundary : Type uBoundary}
  (screenOf : Boundary -> Screen)
  {b1 b2 : Boundary}
  (hscreen : screenOf b1 = screenOf b2)
  (hdistinct : b1 != b2) :
  ! BareScreenSelectsBoundary screenOf := by
  intro hsel
  obtain <select, hselect> := hsel
  apply hdistinct
  calc
    b1 = select (screenOf b1) := (hselect b1).symm
    _ = select (screenOf b2) := congrArg select hscreen
    _ = b2 := hselect b2

/-- A bulk assignment factors through the bare screen when the reconstructed
bulk depends only on bare screen data and on no further boundary-theory data. -/
def BulkFactorsThroughBareScreen
  {Screen : Type uScreen} {Boundary : Type uBoundary} {Bulk : Type uBulk}
  (screenOf : Boundary -> Screen) (bulkOf : Boundary -> Bulk) : Prop :=
  FactorsThrough screenOf bulkOf

/-- T20 (No Bulk From Bare Boundary): two boundary theories with one bare
screen but different bulk reconstruction outputs rule out any reconstruction
map that factors through the bare screen alone. -/
theorem same_screen_different_bulk_refutes_bare_screen_holography
  {Screen : Type uScreen} {Boundary : Type uBoundary} {Bulk : Type uBulk}
  (screenOf : Boundary -> Screen) (bulkOf : Boundary -> Bulk)
  {b1 b2 : Boundary}
  (hscreen : screenOf b1 = screenOf b2)
  (hbulk : bulkOf b1 != bulkOf b2) :
  ! BulkFactorsThroughBareScreen screenOf bulkOf := by
  exact physical_distinction_blocks_public_factorization
    screenOf bulkOf hscreen hbulk

/-- A minimal bare screen signature used only to make the reductio concrete.
The theorem is not about the artistic presentation; it records only the
12/30/20 incidence counts of the bare carrier. -/
structure BareTwelvePortScreen where
  ports : Nat
  edges : Nat
  faces : Nat
  deriving DecidableEq

/-- The named "Mr Tickle Screen" bare carrier used in the reductio. -/
def mrTickleBareScreen : BareTwelvePortScreen :=
  { ports := 12, edges := 30, faces := 20 }

/-- Two different boundary dynamics placed on the same bare twelve-port
carrier. The constructors stand for different physical boundary theories,
not different drawings. -/
inductive MrTickleBoundaryTheory where
  | silent
  | interacting
  deriving DecidableEq

/-- Forget all dynamics and remember only the same 12/30/20 bare screen. -/
def mrTickleScreenOf (_ : MrTickleBoundaryTheory) : BareTwelvePortScreen :=
  mrTickleBareScreen

/-- A toy physical discriminator: the two boundary theories differ. -/
def mrTickleBoundaryReadout : MrTickleBoundaryTheory -> Bool
  | .silent => false
  | .interacting => true

/-- T21 (Mr Tickle Screen reductio): the bare twelve-port screen cannot
recover even this minimal distinction between two boundary theories living on
exactly the same carrier. Therefore incidence alone is not a boundary-QFT
selector. -/
theorem mr_tickle_screen_reductio :
  ! FactorsThrough mrTickleScreenOf mrTickleBoundaryReadout := by
  apply physical_distinction_blocks_public_factorization
    mrTickleScreenOf mrTickleBoundaryReadout
    (p := MrTickleBoundaryTheory.silent)
    (q := MrTickleBoundaryTheory.interacting)
  * rfl
  * decide

/-- Current-public-floor convenience corollary: a family of 8,800
source-factorized judgements still has zero discrimination on a same-reduct

```



```
physical pair. The general theorem remains count-independent. -/  
theorem eight_thousand_eight_hundred_source_theorems_zero_discrimination  
  {Src : Type uSrc} {Phys : Type uPhys}  
  (red : Phys -> Src) (judge : Fin 8800 -> Phys -> Bool)  
  (hfactor : TheoremFamilyFactorsThrough red judge)  
  {p q : Phys} (hred : red p = red q) :  
  ! FamilyDistinguishes judge p q := by  
  exact source_theorem_family_zero_discrimination red judge hfactor hred  
end PhysicalNonDefinability
```

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